AI TRAVEL WATCHDOG & GBV SAFETY COMPANION

Protecting Every Commuter. Zero-Knowledge Safety in 11 Languages.

Phola combines real-time transit anomaly detection, native in-app turn-by-turn navigation to SAPS police stations, 1-tap SOS panic, and encrypted digital evidence preservation.

JUDGE / ASSESSOR TESTING ACCOUNT (LIVE & ACTIVE)

Email: jackympoka@gmail.com | Password: Mpokabetty221@

Live Web App (PWA)

https://phola-beta.vercel.app

Lead Developer: Jacky Mpoka

GitHub: https://github.com/jackyphuti/Phola

PROBLEM STATEMENT

South Africa's Transit & GBV Safety Crisis

Vulnerability after dark, delayed emergency response, and evidence destruction.

Critical Context

1 in 3 Women

South Africa suffers from pandemic-level rates of Gender-Based Violence (GBV). Victims frequently travel alone and lack safe, discreet intervention channels.

Night Transit Risk

Millions of commuters rely on e-hailing (Uber/Bolt) and minibus taxis. Unauthorized route deviations, unregistered drivers, and dark transit corridors pose severe risks.

45+ Min 10111 Delay

Calling 10111 or 112 during an active attack or robbery is dangerous, slow, and fails to transmit real-time satellite GPS coordinates to first responders.

Evidence Destruction

Perpetrators routinely seize, search, or destroy victim devices. Traditional photos and text messages are lost, undermining legal prosecution in courts.

SOLUTION ARCHITECTURE

✨ Core Solution

Introducing Phola: An Autonomous Digital Bodyguard

A multi-layered safety shield designed specifically for South African commuters.



SafeRide Watchdog

Monitors trips autonomously. If your driver deviates from the expected route or stops in an unlit corridor, Phola automatically checks in and pings your emergency circle.



Native Police Navigation

1-tap routing directly inside Phola to the nearest SAPS police station or trauma clinic. Continuous turn-by-turn guidance keeps the user protected inside the app.



Silent 1-Tap SOS

Instant multi-channel panic alerting SAPS (10111), Emergency Medical (112), and the National GBV Command Centre (0800 428 428) with live satellite coordinates.



Zero-Knowledge Vault

Audio notes, photo logs, and incident diaries are encrypted on-device via Web Crypto SHA-256 and AES-GCM before syncing to the cloud. Tamper-evident and secure.



Stealth Decoy Exit

Under physical threat or phone inspection, 1 tap on "Quick Exit" instantly wipes DOM trace states and presents an innocent local weather forecast within 50ms.

ZA

11 Official Languages

Full native accessibility in isiZulu, isiXhosa, Afrikaans, Sepedi, Setswana, Sesotho, Xitsonga, siSwati, Tshivenda, isiNdebele, and English.

USER STORY

Thandi's Midnight Ride: Real-World Experience

How Phola protects a student returning home late from campus.

📖 Persona Journey

11:15 PM — **Booking Ride & Activating Watchdog**

Thandi leaves the university library and requests an e-hailing cab. She taps **"Start SafeRide"** in Phola. The app locks her vehicle notes and initiates background GPS tracking.

11:28 PM — **Route Deviation Anomaly Detected**

The vehicle abruptly takes an unlit access bypass away from her residential address. Phola detects the corridor divergence, alerts Thandi, and silently transmits her live GPS pin to her sister.

11:30 PM — **1-Tap Safe Haven: Go to Nearest SAPS Station**

Feeling threatened, Thandi taps **"Go to Nearest Police Station"**. Phola automatically plots an in-app high-contrast route to Brixton SAPS (1.8 km away) with live turn-by-turn instruction banners.

11:36 PM — **Safe Arrival & Cryptographic Evidence Preserved**

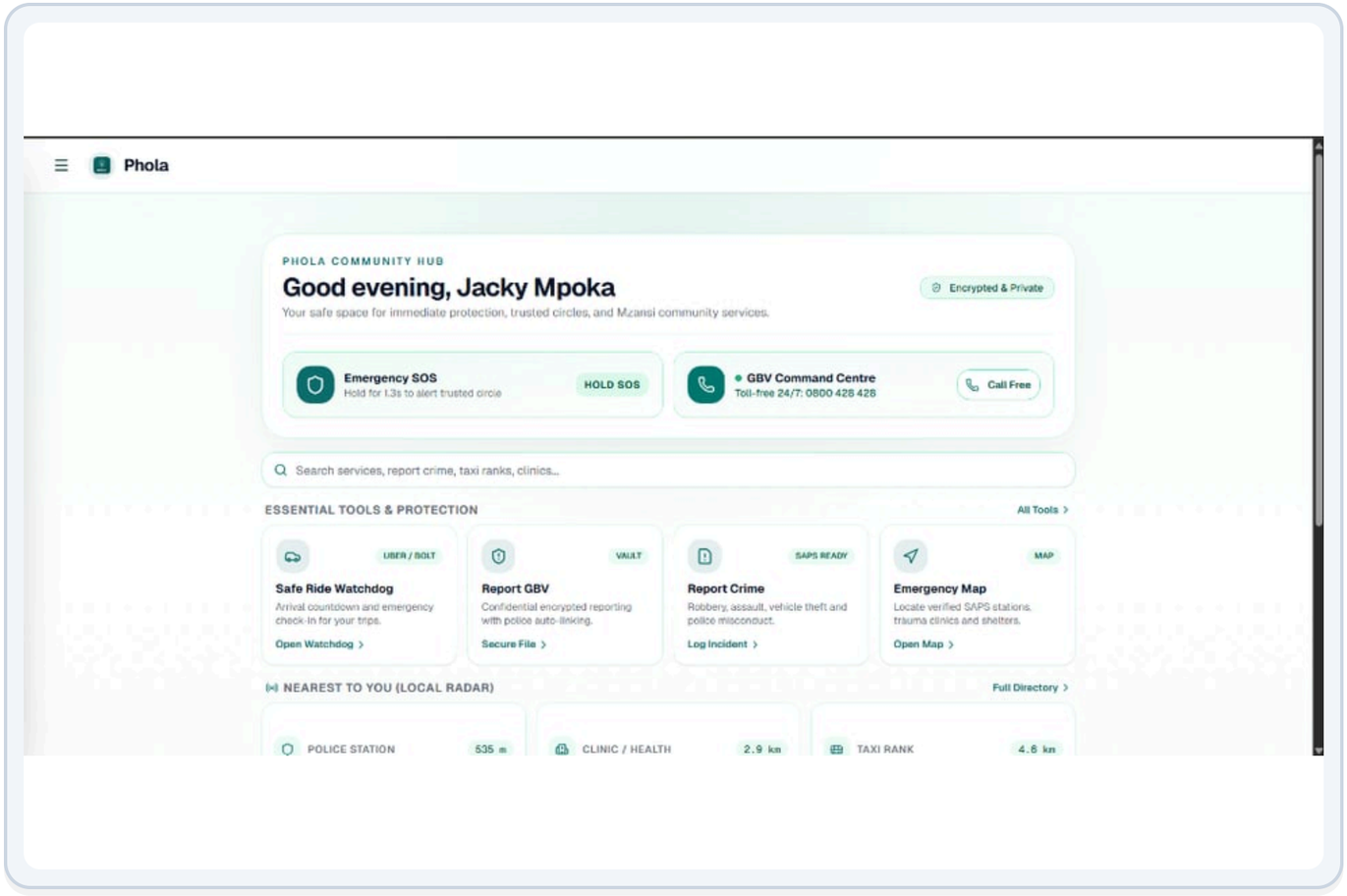
Driver pulls into the illuminated police station gate. Thandi exits safely. The entire telemetry log and route record remain encrypted in her Zero-Knowledge incident vault.

APP WALKTHROUGH 1/8

Community Hub: Live Dashboard & Radar

Central safety hub displaying real-time emergency hotlines and nearest facility radar.

🏠 Home Dashboard



User-Centric Safety

Live Personalized Command Center

Greets the user (**Good evening, Jacky Mpoka**) with encrypted and private status indicators, immediately establishing trust and security.

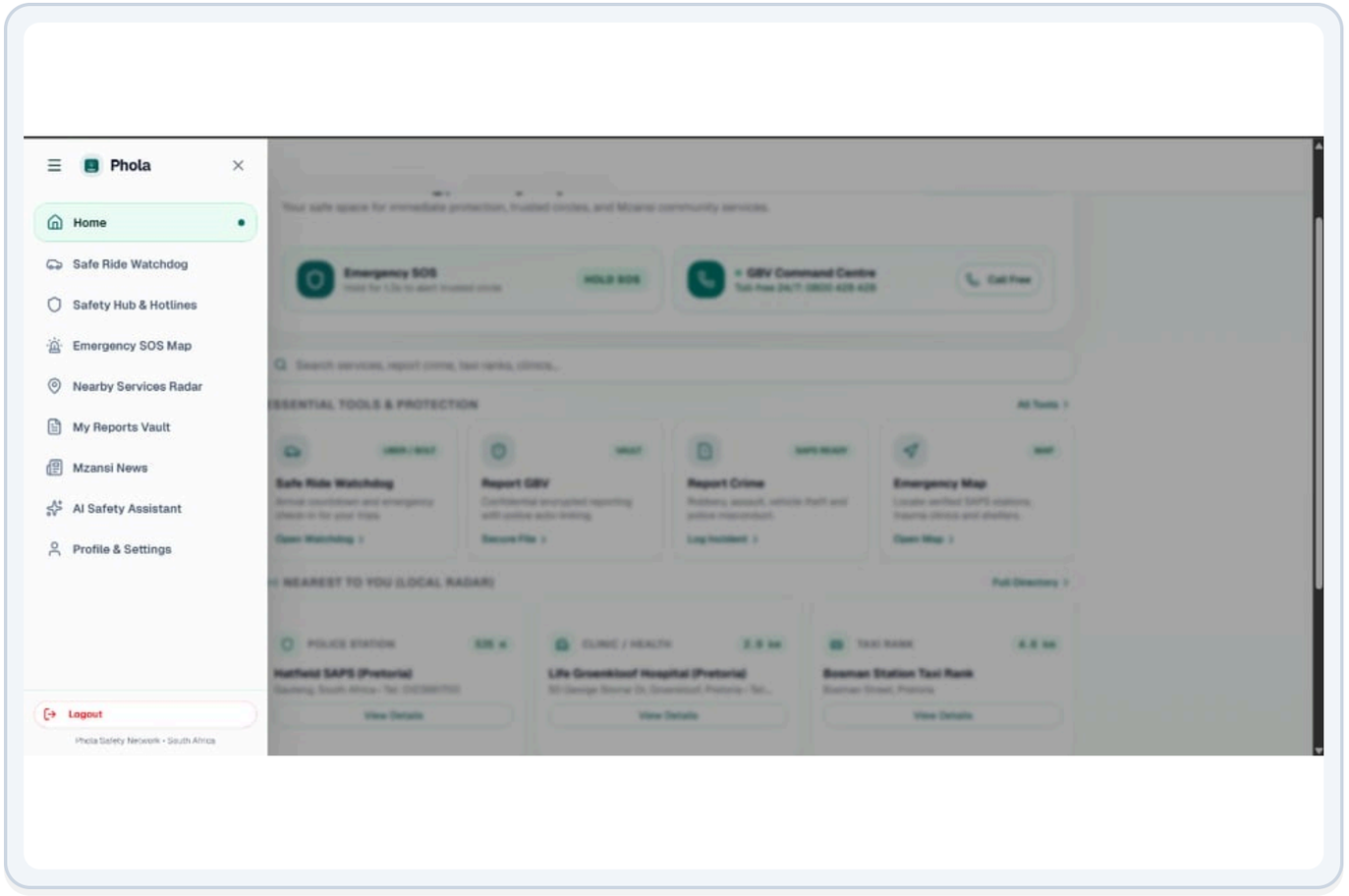
- **1.3s Press-and-Hold SOS:** Prevents accidental pocket triggers while allowing instantaneous activation under duress.
- **National GBV Command Centre:** Direct 1-tap toll-free integration to **0800 428 428** with zero airtime required.
- **Local Services Radar:** Auto-calculates distance to Hatfield SAPS (535m), Life Groenkloof Clinic (2.9km), and Bosman Taxi Rank (4.6km).

APP WALKTHROUGH 2/8

Modular Architecture & Navigation Drawer

Clean, rapid access to all emergency modules, evidence vault, and settings.

Navigation Drawer



System Layout

Nine Core Safety Modules

Designed for high-contrast accessibility. Users can switch between active transit protection and background configuration in one tap.

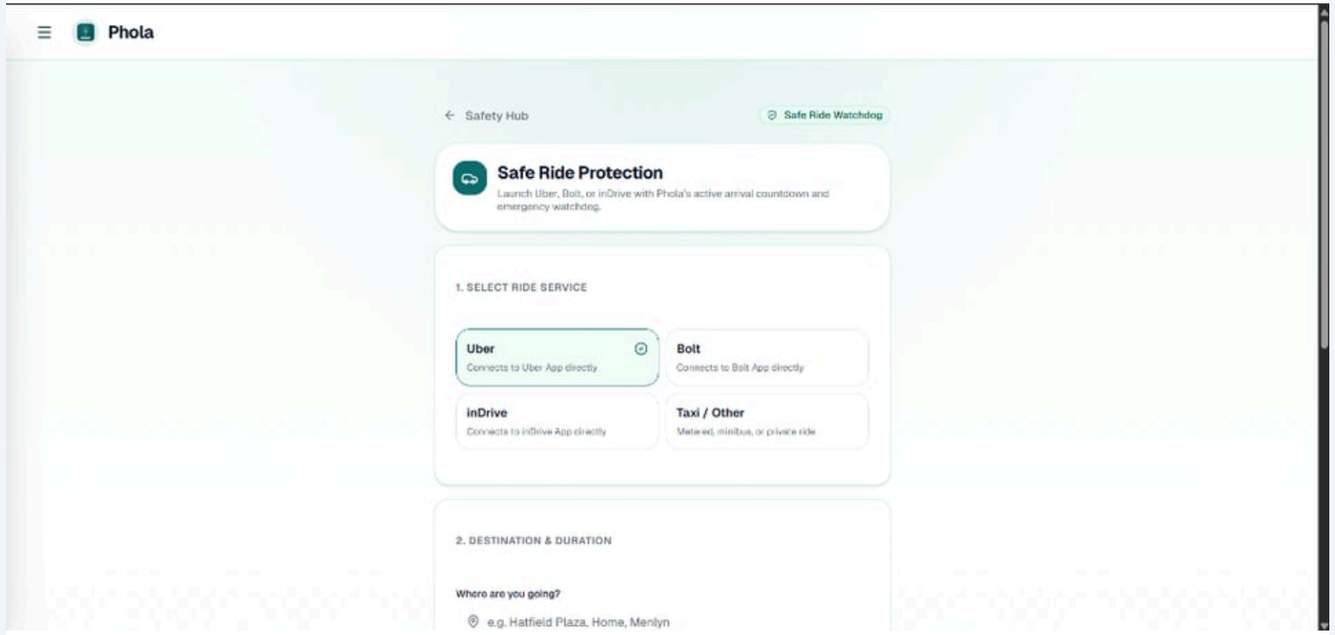
- **Safe Ride Watchdog:** Dedicated trip companion and corridor tracking.
- **Emergency SOS Map:** Fullscreen live Mapbox spatial locator with 35 safe havens.
- **AI Safety Assistant (Gemini):** Discrete guidance powered by Google Gemini.
- **Unblocked Clean Logout:** Users can sign out immediately without any PIN hurdles.

APP WALKTHROUGH 3/8

Safe Ride Watchdog: Multi-Modal Transit Security

Protects commuters across e-hailing apps, metered cabs, and minibus taxis.

🚕 Safe Ride Protection



Universal Coverage

Works with Any South African Transit Mode

Passengers select their provider (**Uber, Bolt, inDrive, or Taxi / Other**) to establish automated watchdog monitoring.

⏱️ **Arrival Countdown:** Real-time estimation calculates corridor duration. If the ride runs overtime without arrival confirmation, check-in alerts trigger.

📍 **Destination Locking:** Commuter specifies target destination (e.g., Hatfield Plaza, Home, Menlyn) for corridor boundary enforcement.

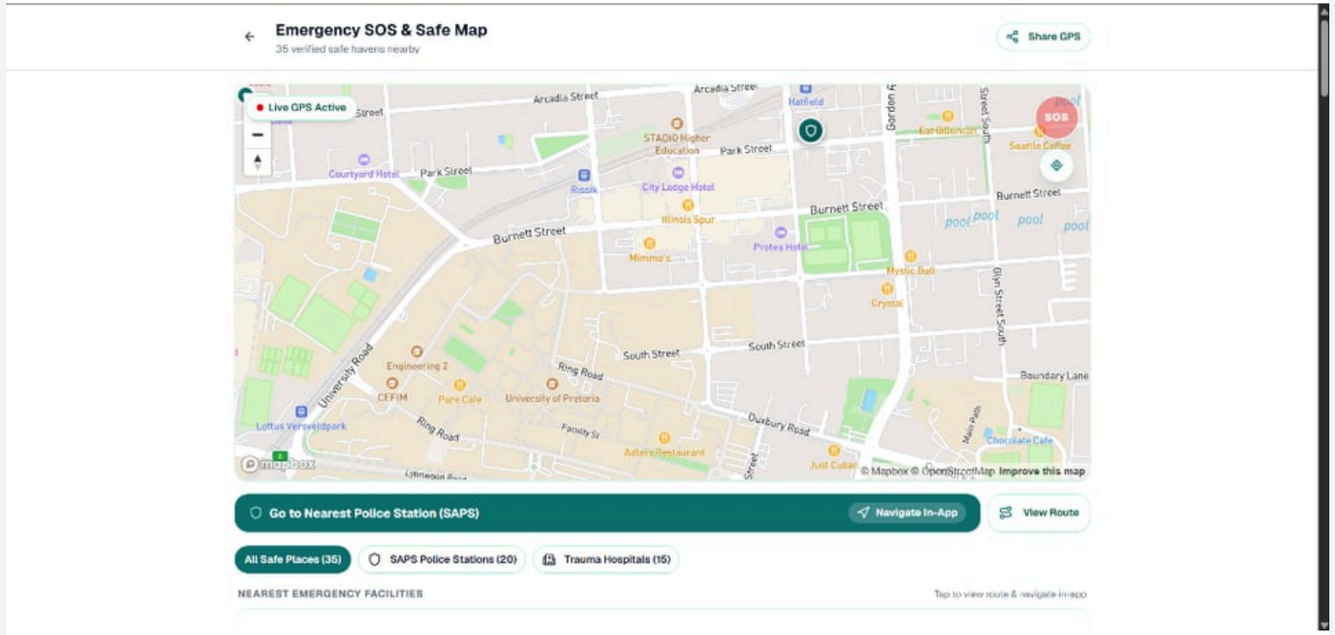
🔄 **Direct App Linking:** Bridges directly to the ride app while maintaining Phola's protective watchdog running in the background.

APP WALKTHROUGH 4/8

Emergency SOS Map: 35 Verified Safe Havens

Real-time GPS tracking with verified police stations and trauma centers in Pretoria.

Spatial Radar



GIS Intelligence

Live Verified Facilities Network

Mapbox GL engine plots **35 verified safe places**: 20 SAPS Police Stations and 15 Trauma Hospitals with live geospatial distance calculations.

Live GPS Active: Continuous satellite positioning tracking user location, speed, and heading vector.

1-Tap Police Bar: Prominent emerald banner: "*Go to Nearest Police Station (SAPS)*" for instantaneous evasion.

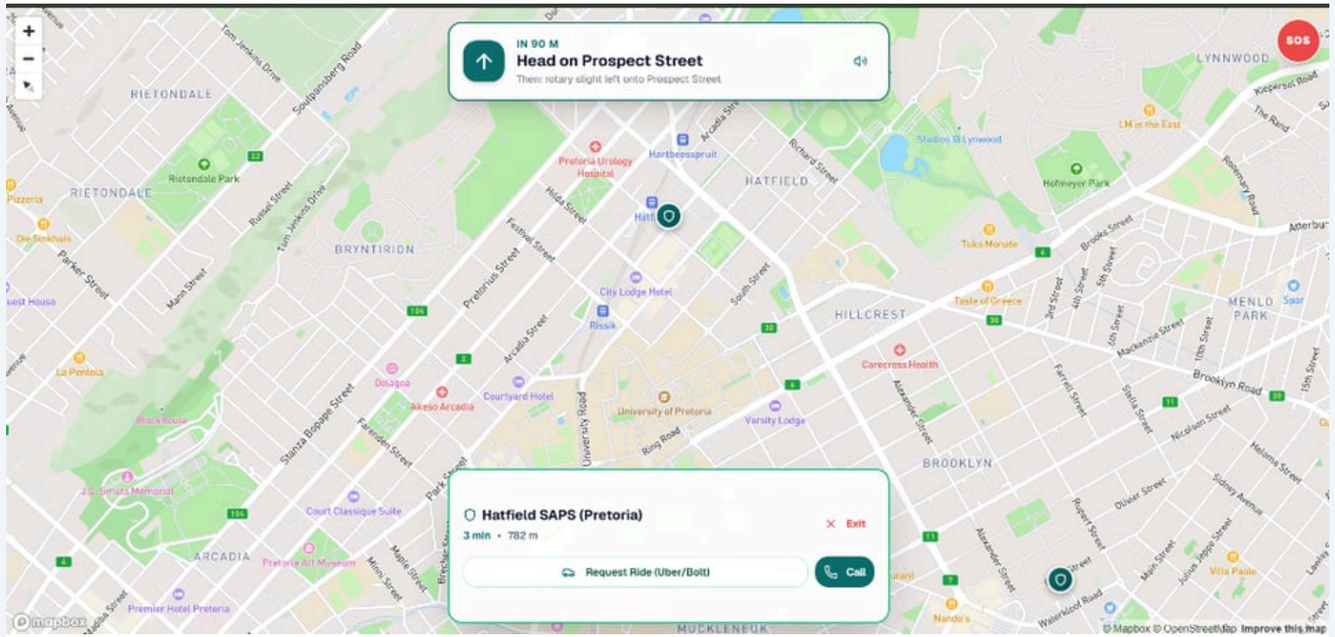
Share GPS: Instant satellite location broadcast via WhatsApp or SMS to emergency trusted contacts.

APP WALKTHROUGH 5/8 • BREAKTHROUGH FEATURE

Live In-App Navigation: 100% Inside Phola

Turn-by-turn routing directly to Hatfield SAPS without relying on external map apps.

In-App GPS Routing



Breakthrough Innovation

Zero App-Switching Security

Traditional panic apps switch to Google Maps, which an attacker can easily close. Phola executes **full turn-by-turn GPS navigation entirely inside its secure container**.

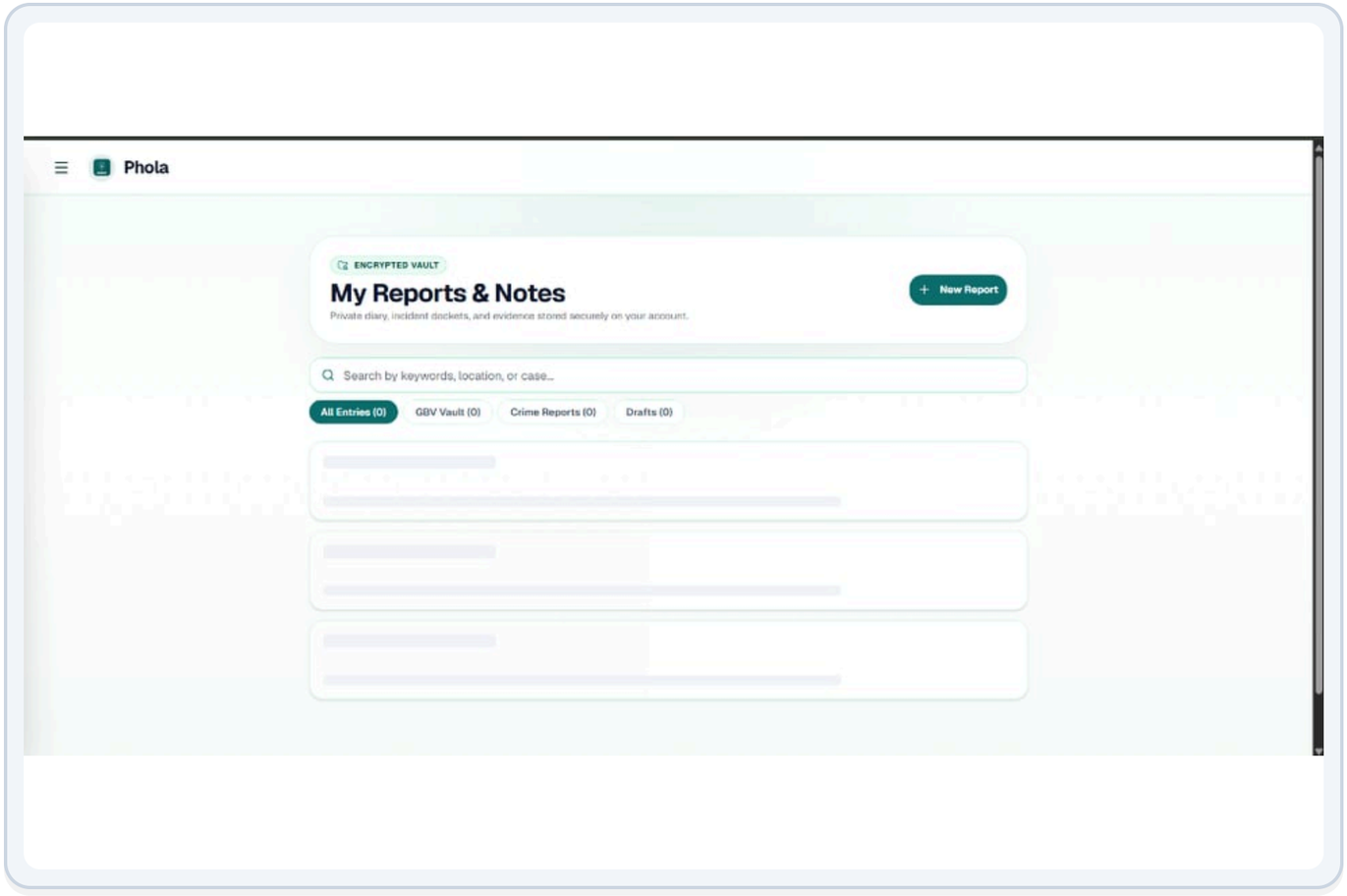
- **Real-Time Turn Guidance:** "*IN 90 M: Head on Prospect Street, then rotary slight left onto Prospect Street*" with voice instruction support.
- **Precision ETA & Distance:** Hatfield SAPS (Pretoria), **3 min • 782 m** with live rerouting.
- **Instant Action Controls:** Integrated "*Request Ride (Uber/Bolt)*", "*Call Station*", and "*Exit*" buttons.

APP WALKTHROUGH 6/8

My Reports: Zero-Knowledge Incident Vault

Client-side encrypted diary and evidence repository complying with POPIA.

🔒 Encrypted Vault



Legal & Forensic Security

Tamper-Proof Evidence Preservation

Protects sensitive incident dockets, voice recordings, and injury photos. Encrypted on the device using **Web Crypto AES-GCM** before transmission.

🗂️ **Segregated Category Vaults:** Fast filtering across *All Entries, GBV Vault, Crime Reports*, and *Drafts*.

🔍 **Encrypted Search:** Filter records by keywords, location coordinates, or SAPS docket case numbers.

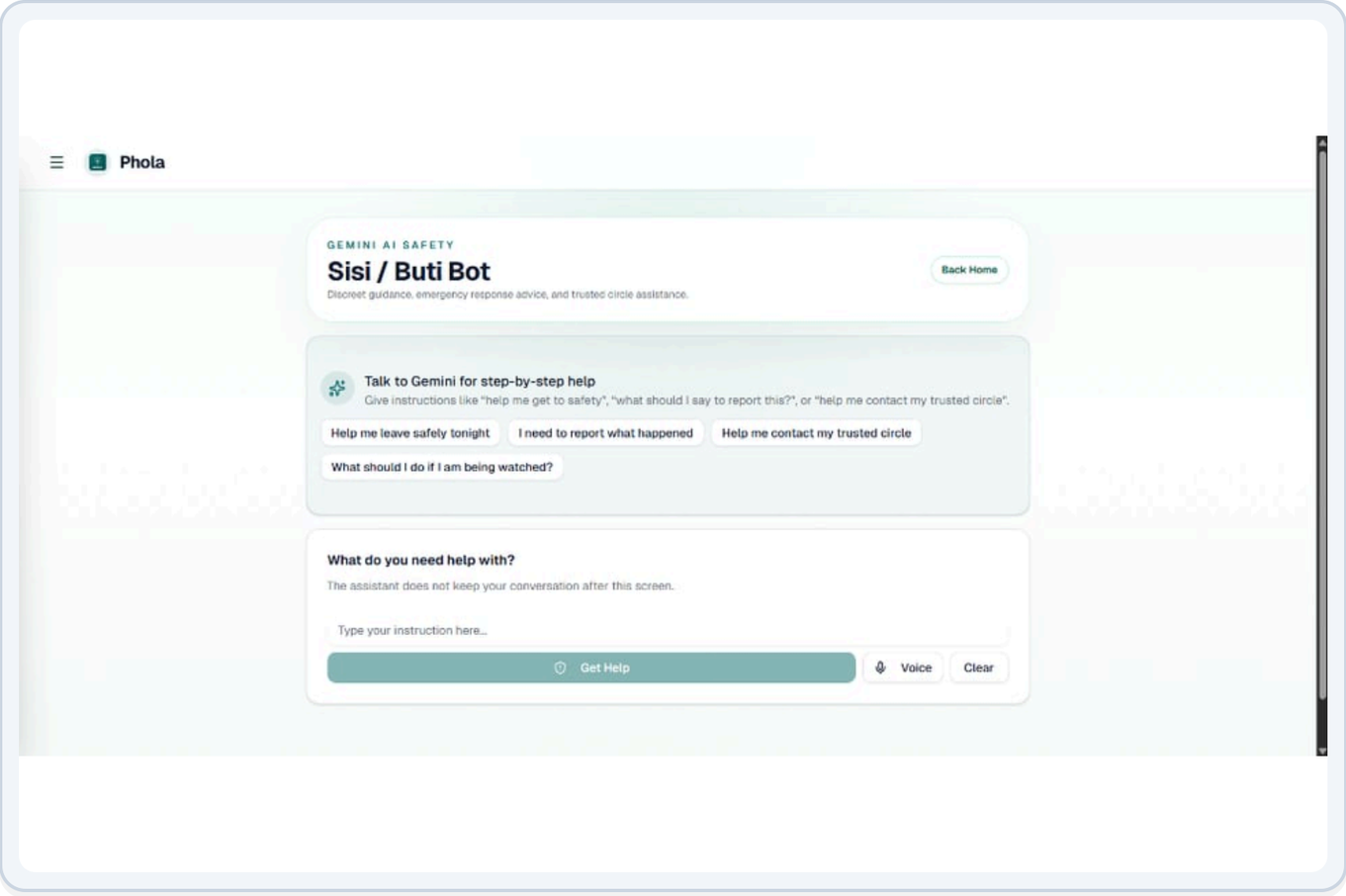
⚖️ **Court-Admissible Metadata:** Records cryptographic timestamps and GPS hashes for South African legal proceedings.

APP WALKTHROUGH 7/8

Sisi / Buti Bot: Gemini AI Safety Assistant

Context-aware, discreet safety guidance powered by Google Gemini AI.

Gemini AI Safety



AI for Public Good

Discreet Emergency Guidance

Named after South African kinship terms (**Sisi / Buti**), providing empathetic, culturally grounded advice during frightening crises.

1-Tap Pre-Configured Prompts: *"Help me leave safely tonight", "I need to report what happened", "Help me contact my trusted circle", "What should I do if I am being watched?"*

Zero Conversation Retention: Explicit safety banner guarantees *"The assistant does not keep your conversation after this screen"* to prevent coercive discovery.

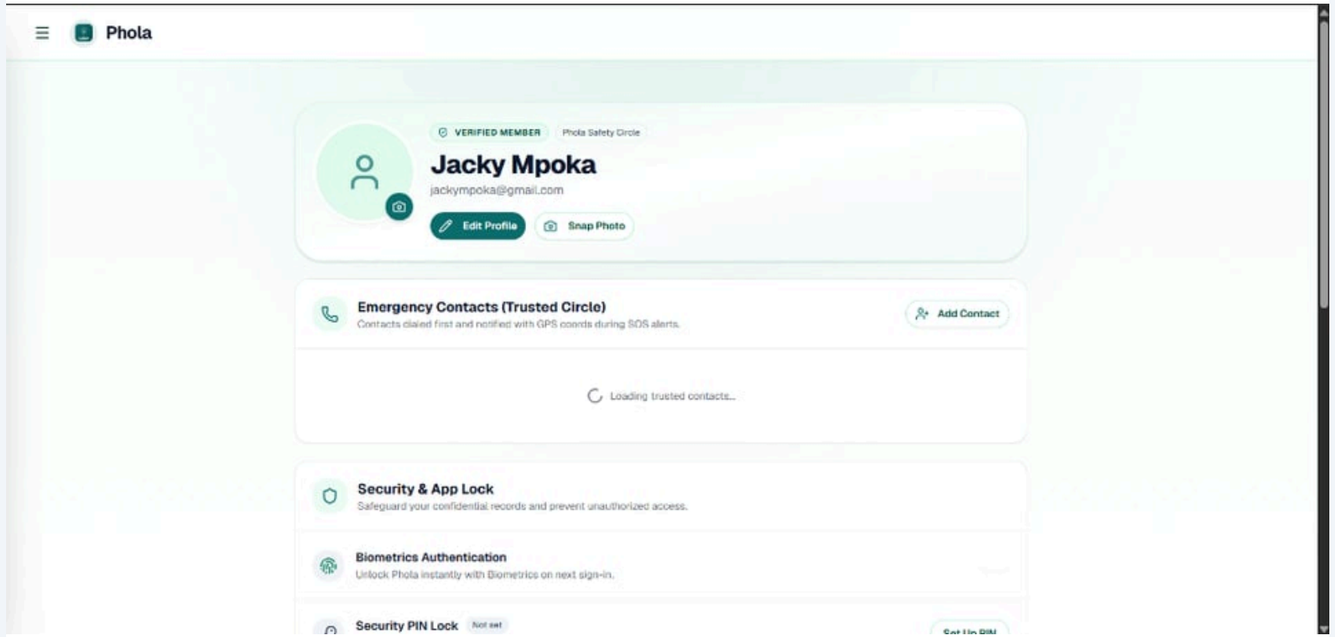
Voice Interaction: Enables hands-free audio querying when typing is too dangerous or physically impossible.

APP WALKTHROUGH 8/8

Profile, Trusted Circle & Biometric Lock

Hardware-backed authentication and POPIA-compliant trusted circle management.

Security Settings



Judge Account Live

Verified Member: Jacky Mpoka

Displays the active demo profile (**jackympoka@gmail.com**) with emergency circle contacts and biometric hardware integration.

👥 **Trusted Circle Contacts:** Configurable emergency contacts dialed first and notified with live GPS coordinates during SOS.

👆 **Biometrics Authentication:** Toggle hardware-level **Face ID on iOS** and **Fingerprint on Android** via WebAuthn standards.

🔢 **Security PIN Lock:** 4-6 digit salted SHA-256 PIN fallback to guard sensitive incident notes.

TECHNICAL ARCHITECTURE

System Design

Production Technology Stack & Resilience

Modern, hybrid, and fault-tolerant architecture built for offline durability.

Frontend & Hybrid

Next.js 16 + Capacitor 5

- React 19 with Turbopack for sub-second build and execution.
- Capacitor 5 Android & iOS bridges compile native native packages.
- Tailwind CSS + Radix UI with South African design palette (`#0D6E6E`).
- Progressive Web App (PWA) with offline caching.

Backend & Storage

Supabase PostgreSQL

- Row-Level Security (RLS) enforcing strict user-isolated boundaries.
- Supabase Auth JWT with automatic stale token invalidation.
- Cloudflare Workers edge proxying for low-latency dispatch.
- Realtime subscriptions for trip tracking.

GIS & Routing

Resilient Geolocation Engine

- Continuous GPS heading & speed telemetry via Geolocation API.
- Mapbox Directions API with automatic fallback to OSRM nodes.
- Offline straight-line compass fallback if network drops completely.
- Turf.js spatial calculation for nearest police station discovery.

Zero External Map Dependency

Unlike competing panic buttons that divert to Google Maps, Phola navigates entirely within the secure app container.

Containerized Safety

SECURITY & COMPLIANCE

SSDLC Certified

Secure SDLC & POPIA Compliance

Adherence to South African Privacy Law and Defense-in-Depth Cryptography.

1. Requirements

• Data Minimization: Zero passive background tracking without active trip.
• Explicit POPIA consent gathered on sign-up.
• Right to be Forgotten: 1-tap complete data purge in Settings.

2. Architecture

• STRIDE Threat Modeling applied across all data flows.
• Zero-Knowledge cryptography: Web Crypto AES-GCM client-side encryption.
• Row-Level Security (RLS) on all database tables.

3. Implementation

• Salted SHA-256 PIN hashing (`${userId}:phola-pin-salt:${pin}`).
• WebAuthn Passkeys & Biometrics (`PublicKeyCredential`).
• Strict TypeScript with zero any-casts in crypto routines.

4. Verification

• Automated SAST & type-checking (`tsc --noEmit`).
• Next.js static page bundle auditing (28 routes verified).
• Native Android APK compiled with Gradle 8 and Java 17.

BUSINESS MODEL

Lean Business Canvas Overview

Viable, scalable, and socially impactful business strategy.

💼 Strategy

Problem & Segments	Value Proposition & Solution	Channels & Advantage	Revenue Streams	Cost Structure
Problem: • 1 in 3 women face GBV • Night e-hailing risks • 45+ min 10111 response delay • Evidence destruction **Customers:** • Commuters & students • Night healthcare staff • Campuses & CPFs	**Unique Value Prop:** "Your autonomous travel watchdog and GBV companion with in-app police navigation and zero-knowledge vaults." **Solution:** • SafeRide trip watchdog • In-app turn-by-turn routing • 1-Tap SOS to 10111/112 • Stealth weather exit • 11 official SA languages	**Channels:** • PWA & Play Store • University campus drives • GBV shelter networks (POWA) • Community policing (CPF) **Unfair Advantage:** • Native in-app navigation • Zero-Knowledge crypto • 11 Official SA languages	• **Free Tier:** Public safety & SOS (always free) • **Phola Plus:** R39/mo armed response integration • **B2B Campus:** R15/student/mo campus safety portal • **CSI & Grants:** GBV justice sponsorships	• Next.js hosting & Supabase DB • Mapbox & OSRM routing APIs • SMS/WhatsApp emergency gateways • POPIA audits & maintenance

JUDGE VERIFICATION

Official Submission Deliverables & Evaluation Guide

All materials required for judging are live, verified, and ready to inspect.

Verified Links

🌐 Live Web Application (PWA)

Hosted and deployed live on Vercel edge infrastructure:

```
https://phola-beta.vercel.app
```

Compatible with modern mobile and desktop browsers. Supports progressive installation and WebAuthn biometrics.

📦 Android APK Package

Compiled native debug APK ready for Android testing:

```
phola-debug.apk (3.64 MB)
```

Built with Gradle 8 & Java 17. Stored directly in the project root directory and available for immediate installation.

💻 Open Source Git Repository

Complete clean commit history on branch `main`:

```
https://github.com/jackyphuti/Phola
```

Includes full source code, Next.js routes, Capacitor mobile bridges, and comprehensive documentation in `PHOLA_SUBMISSION_PACK.md`.

🔑 Judge Demo Credentials

Live active account created for the evaluation panel:

```
Email: jackympoka@gmail.com
Password: Mpokabetty221@
```

Video demo will be uploaded to Google Drive following physical recording.